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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Swetha Nagasri
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Roll Number	2520030094
Branch	CSE
Course Outcome	CO-4

Case Study Topic : Pune–Mumbai Expressway Dynamic Toll Path Optimization using Dijkstra's Algorithm

Work Done Summary :

In this case study, Dijkstra's Algorithm was implemented to determine the minimum travel-cost route between Pune and Mumbai on a directed weighted graph representing the Pune-Mumbai Expressway toll network.

The graph consists of toll plazas as vertices and road segments as weighted edges. Each edge weight represents travel cost, toll charge, or travel time. A priority queue (min-heap) was used to efficiently select the next closest vertex during computation.

The algorithm successfully identified the shortest path from KTR (Pune Entry) to PAN (Mumbai Exit) and demonstrated how dynamic updates to toll charges affect route selection. This approach is useful in intelligent transportation systems and navigation applications.

Implementation Details:

1. Represented toll plazas as graph vertices.
2. Modeled road segments as weighted directed edges.
3. Used adjacency lists to store graph connections.
4. Initialized all distances to infinity except the source vertex.
5. Applied Dijkstra's Algorithm using a priority queue.
6. Updated shortest distances through edge relaxation.
7. Maintained predecessor information for path reconstruction.
8. Generated the shortest route from Pune to Mumbai.

9. Simulated dynamic toll updates by modifying edge weights.
10. Recomputed shortest paths after updates.

Result: Output Screenshots:

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PUNE-MUMBAI EXPRESSWAY ROUTE OPTIMIZATION
Using Dijkstra's Algorithm
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Source Toll Plaza : KTR
Destination Toll Plaza : PAN

Computing shortest path...

Visiting: KTR
Updating distance of LNB = 12
Updating distance of KDA = 15

Visiting: LNB
Updating distance of TLG = 22

Visiting: KDA
Updating distance of KDB = 31

Visiting: TLG
Updating distance of DNB = 42

Visiting: KDB
No shorter path found for DNB

Visiting: DNB
Updating distance of PAN = 52
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========================================
SHORTEST PATH FOUND
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KTR -> LNB -> TLG -> DNB -> PAN

Minimum Cost = 52

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DISTANCE TABLE
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Vertex      Distance From KTR
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KTR         0
LNB         12
KDA         15
TLG         22
KDB         31
DNB         42
PAN         52

========================================
PATH RECONSTRUCTION
========================================

PAN <- DNB <- TLG <- LNB <- KTR

Final Route:
KTR -> LNB -> TLG -> DNB -> PAN

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EXECUTION COMPLETED SUCCESSFULLY
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Conclusion :

The Dijkstra-based route optimization algorithm successfully identified the minimum-cost route from Pune to Mumbai on the expressway toll network. The algorithm efficiently handled weighted road segments and produced the optimal path using a priority queue. Dynamic toll updates demonstrated how shortest routes can change over time, making this approach highly suitable for modern GPS navigation, transportation management, and smart traffic systems.

Student Signature	**Faculty Signature**